



UNSW
SYDNEY

MATH2601 – Higher Linear Algebra

Topic 1 – Groups and fields

Lecture 1.1 – Groups and subgroups

Lecturer: Dr Sean Gardiner – sean.gardiner@unsw.edu.au

Introduction

In MATH1141, we defined geometric vectors, and associated them with real column vectors in $\mathbb{R}^2$, $\mathbb{R}^3$, etc. Then in MATH1241, we generalised the notion of vectors as elements of vector spaces, treating other objects like matrices over $\mathbb{R}$ or $\mathbb{C}$ and polynomials as vectors. We also encountered the idea of a linear transformation, and found a correspondence between linear transformations and matrices. Through this, we saw a way to diagonalise certain types of matrices and some applications of diagonalisation.

In MATH2601, we will expand upon these generalisations by taking a step further into the abstract, defining the notion of an algebraic group and using that definition to better understand the structure of vector spaces and their scalar fields (Topics 1 and 2). We will also generalise some geometric properties associated with vectors (Topic 4) and some inherent properties of the matrix determinant (Topic 5).

We will further generalise the notion of a linear transformation and its correspondence with matrices (Topic 3), and expand upon the ideas underlying diagonalisation in order to find a more general form that any type of matrix can be associated with (Topics 6 and 7). Using these ideas, some other applications will be explored in Topic 8.

Groups

Definition. A **group** is a set G equipped with a binary operator $*$ that satisfies the following axioms:

- For all $a, b \in G$, we have $a * b \in G$. (**Closure.**)
- For all $a, b, c \in G$, we have $a * (b * c) = (a * b) * c$. (**Associativity.**)
- There exists an element $e \in G$ for which $a * e = e * a = a$ for all $a \in G$. (Existence of an **identity.**)
- For each $a \in G$, there exists an $a' \in G$ for which $a * a' = a' * a = e$. (Existence of **inverses.**)

Notation. Since the definition of a group is in terms of both a set G and an operation $*$ acting on it, a group can also be written as the pair $(G, *)$.

Notice that a group must always be non-empty, due to the existence of an identity axiom. If a finite group has n elements, we say it has **order** n .

Example. The **trivial group** is the group $(\{e\}, *)$, where e is necessarily the group's identity element.

Definition. A group $(G, *)$ is called **abelian** if it also satisfies the axiom:

- For all $a, b \in G$, we have $a * b = b * a$. (**Commutativity.**)

Group operations

In the context of groups $(G, *)$, often the operation $*$ ends up being a form of addition, represented by the symbol $+$, or multiplication, represented by the symbol $\times$ (or just by **concatenation**, writing ab instead of $a \times b$).

When the operation is $+$, we have the following shorthand notation:

- The (additive) identity is denoted by 0 .
- The (additive) inverse of an element $a \in G$ is denoted by $-a$.
- Subtraction between elements $a, b \in G$ is defined as $a - b = a + (-b)$.
- Repeated application of the operation is denoted by

$$\underbrace{a + a + \cdots + a}_{N \text{ copies}} = Na.$$

When the operation is $\times$, we have the following shorthand notation:

- The (multiplicative) identity is denoted by 1 .
- The (multiplicative) inverse of an element $a \in G$ is denoted by a^{-1} .
- Division between elements $a, b \in G$ is defined as $a/b = a \times b^{-1} = ab^{-1}$.
- Repeated application of the operation is denoted by $\underbrace{a \times a \times \cdots \times a}_{N \text{ copies}} = a^N$.

Examples of groups

Exercise. Which of the following are groups?

- $(\mathbb{Z}, +)$ **is** a group, because it satisfies the four group axioms.
- $(\mathbb{Z}^+, +)$ is **not** a group, because it doesn't satisfy the identity axiom (specifically $0 \notin \mathbb{Z}^+$) (nor the inverses axiom).
- $(\mathbb{N}, +)$ is **not** a group, because it doesn't satisfy the inverses axiom (e.g. $1 \in \mathbb{N}$ but $-1 \notin \mathbb{N}$).
- $(\{-1, 1\}, \times)$ **is** a group, because it satisfies the four group axioms.
- $(\mathbb{Z}, \times)$ is **not** a group, because it doesn't satisfy the inverses axiom (e.g. $2 \in \mathbb{Z}$ but 2 has no multiplicative inverse in $\mathbb{Z}$).
- $(\mathbb{Q}, \times)$ is **not** a group, because it doesn't satisfy the inverses axiom (specifically, $0 \in \mathbb{Q}$ but 0 has no multiplicative inverse in $\mathbb{Q}$).
- $(\mathbb{Q} - \{0\}, \times)$ **is** a group, because it satisfies the four group axioms.
- $(\mathbb{R}^+, \times)$ **is** a group, because it satisfies the four group axioms.
- $(\mathbb{R}^-, \times)$ is **not** a group, because it doesn't satisfy the closure axiom (e.g. $-1, -2 \in \mathbb{R}^-$ but $-1 \times -2 = 2 \notin \mathbb{R}^-$) (nor the identity axiom).

Note that all the examples above that are groups are in fact **abelian** groups.

The set of integers modulo m

Definition. Given any positive integer m , we can define a number system $\mathbb{Z}_m = \{0, 1, 2, \dots, m-1\}$ that respects usual addition and multiplication modulo m , that is, addition and multiplication are the same as in $\mathbb{Z}$, except always reduced to the smallest non-negative remainder when divided by m .

For example, in $\mathbb{Z}_7$, we have $2 + 4 = 6$, $3 + 4 = 0$, and $4 + 5 = 2$.

Similarly, in $\mathbb{Z}_8$, we have $2 \times 3 = 6$, $2 \times 4 = 0$, and $3 \times 5 = 7$.

Exercise. Show that $(\mathbb{Z}_4, +)$ is a group.

Solution. By definition, $\mathbb{Z}_4$ is closed under addition.

The associativity of addition in $\mathbb{Z}_4$ is inherited from the operation in $\mathbb{Z}$.

The identity in $(\mathbb{Z}_4, +)$ is the element 0, since for all $a \in \mathbb{Z}_4$, we have $a + 0 = 0 + a = a$.

For each element in $\mathbb{Z}_4$, an inverse exists: $-0 = 0 \in \mathbb{Z}_4$, $-1 = 3 \in \mathbb{Z}_4$, $-2 = 2 \in \mathbb{Z}_4$, and $-3 = 1 \in \mathbb{Z}_4$.

So $(\mathbb{Z}_4, +)$ is a group.

The group is also abelian since the commutativity of addition is inherited from $\mathbb{Z}$.

The set of integers modulo m

In general, for sets $\mathbb{Z}_m$, we are guaranteed closure by definition, and guaranteed associativity and commutativity of $+$ or $\times$ as inherited from $\mathbb{Z}$.

Exercise. Which of the following are groups?

- $(\mathbb{Z}_m, +)$ for any $m \in \mathbb{Z}^+$ is a group. The identity is 0, and the inverse of 0 is itself, while the inverse of any nonzero $a \in \mathbb{Z}_m$ is $m - a \in \mathbb{Z}_m$.
- $(\mathbb{Z}_m, \times)$ for any integer $m > 1$ is **not** a group, because it doesn't satisfy the inverses axiom ($0 \in \mathbb{Z}_m$ but 0 has no multiplicative inverse in $\mathbb{Z}_m$).
- $(\mathbb{Z}_4 - \{0\}, \times)$ is **not** a group, because it doesn't satisfy the inverses axiom. Specifically, $2 \in \mathbb{Z}_4 - \{0\}$ but 2 has no inverse in $\mathbb{Z}_4 - \{0\}$.
- $(\mathbb{Z}_5 - \{0\}, \times)$ is a group. The identity is 1, and each element has an inverse since $1^{-1} = 1$, $2^{-1} = 3$, $3^{-1} = 2$, and $4^{-1} = 4$.
- $(\mathbb{Z}_p - \{0\}, \times)$ for any **prime** number p is a group. In particular, each element must have an inverse since it is coprime with p (MATH1081).
- $(\mathbb{Z}_m - \{0\}, \times)$ for any **non-prime** $m \in \mathbb{Z}^+$ is **not** a group, because it doesn't satisfy the closure axiom. Specifically, $m = ab$ for some integers $a, b > 1$, so $ab = 0$ under $\times$, but 0 is not included in the group.

Each of the examples above that are groups are again **abelian**.

Matrix groups

Recall that notation like $M_{p,q}(\mathbb{R})$ represents the set of all $p \times q$ matrices with real entries, and $M_n(\mathbb{C})$ is the set of all $n \times n$ matrices with complex entries.

Example. The set $M_{p,q}(\mathbb{R})$ is a group under matrix addition. Notice that in this case, the identity element is $\mathbf{0}_{p,q}$, the $p \times q$ zero matrix.

Exercise. Explain why the set $M_n(\mathbb{R}) - \mathbf{0}_n$ is **not** a group under matrix multiplication for $n > 1$.

Solution. Notice that in this case, the identity element is I_n , the $n \times n$ identity matrix. But there exist non-zero $n \times n$ real matrices A that are not invertible (e.g. any such matrix with a zero row), meaning there is no matrix B for which $AB = I$, and so the existence of inverses axiom is not satisfied.

Definition. The set of all invertible $n \times n$ matrices with real entries is called the **general linear group** of degree n over $\mathbb{R}$, denoted $\text{GL}(n, \mathbb{R})$.

Example. The set $\text{GL}(n, \mathbb{R})$ is a group under matrix multiplication. It is abelian if and only if $n = 1$.

All of the above is also true if we replace $\mathbb{R}$ with $\mathbb{C}$, or indeed with any field $\mathbb{F}$ (to be defined in Lecture 1.2).

Group of bijective functions

Exercise. Let X be a set and $G = \{\text{bijective functions } f : X \rightarrow X\}$. Show that G is a group under function composition $\circ$.

Solution. Since the composition of bijective functions is also bijective, we know G satisfies the **closure** axiom.

For all $f, g, h \in G$, for any $x \in X$, we have

$$((f \circ g) \circ h)(x) = (f(g(h(x)))) = (f(g(h(x)))) = (f \circ (g \circ h))(x).$$

So G satisfies the **associativity** axiom.

Since the identity function $\text{id} : X \rightarrow X$ is a bijection, we know $\text{id} \in G$ is the **identity** element of G since for all $f \in G$, for any $x \in X$, we have

$$(f \circ \text{id})(x) = f(\text{id}(x)) = f(x) = \text{id}(f(x)) = (\text{id} \circ f)(x).$$

Finally, for all $f \in G$, since f is bijective we know that its inverse $f^{-1} : X \rightarrow X$ exists and is also bijective, so $f^{-1} \in G$ meaning G satisfies the existence of **inverses** axiom.

So G is a **group** under function composition.

Note that in this case, G is **not** abelian in general.

Checking finite groups

If we are asked to check whether a small finite set is a group, it is often most efficient to just calculate all possible operation results using a table.

Exercise. Show that $G = \{1, -1, i, -i\}$ is a group under multiplication.

Solution. We first draw up a multiplication table for the four elements:

$\times$	1	-1	i	$-i$
1	1	-1	i	$-i$
-1	-1	1	$-i$	i
i	i	$-i$	-1	1
$-i$	$-i$	i	1	-1

We can see that G is **closed** under multiplication since all entries in the table are elements of G .

We can also see from the table that 1 is the **identity** element.

The table also shows that every element has an **inverse**, since the identity element 1 appears in every row (for example, $i^{-1} = -i$ since $i(-i) = 1$.)

For **associativity**, we can simply state that the associativity of multiplication in G is inherited from multiplication in $\mathbb{C}$.

Since it satisfies all four group axioms, G is a **group** under multiplication.

Since the multiplication table is symmetric, we further see that G is **abelian**.

Checking finite groups

Let $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, $A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, $B = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$, $C = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$.

Exercise. Show that $G = \{I, A, B, C\}$ is a group under multiplication.

Solution. We first draw up a multiplication table for the four elements:

$\times$	I	A	B	C
I	I	A	B	C
A	A	I	C	B
B	B	C	I	A
C	C	B	A	I

We can see that G is **closed** under multiplication since all entries in the table are elements of G .

We can also see from the table that I is the **identity** element.

The table also shows that every element has an **inverse**, since the identity element I appears in every row (for example, $A^{-1} = A$ since $AA = I$.)

For **associativity**, we can simply state that the associativity of multiplication in G is inherited from matrix multiplication in $M_2(\mathbb{Z})$.

Since it satisfies all four group axioms, G is a **group** under multiplication.

Since the multiplication table is symmetric, we further see that G is **abelian**.

Properties of groups

Let $(G, *)$ be a group. Then the following properties hold:

Lemma. There is only one identity element in G .

Proof. Suppose both e and e' are identity elements in G . Then $e * e' = e'$ since e is an identity element, but also $e * e' = e$ since e' is an identity element. So in fact $e = e'$ and the identity element must be unique.

Lemma. Each element in G has a unique inverse.

Proof. Let $a \in G$, and suppose both a' and a'' are inverses of a in G . Then $(a' * a) * a'' = e * a'' = a''$ since a' is an inverse of a , but also $(a' * a) * a'' = a' * (a * a'') = a' * e = a'$ since a'' is an inverse of a (and G is associative). So in fact $a' = a''$ and the inverse of a must be unique.

Lemma. For each $a \in G$, we have $(a^{-1})^{-1} = a$.

Proof. Let $a \in G$. The inverse of a^{-1} is the element $b \in G$ satisfying $a^{-1} * b = e$. We know that $a^{-1} * a = e$ since a^{-1} is the inverse of a . Since inverses are unique, this must mean that $b = a$, that is, $(a^{-1})^{-1} = a$.

Properties of groups

Let G be a group under multiplication (represented as concatenation). Then:

Lemma. For each $a, b \in G$, we have $(ab)^{-1} = b^{-1}a^{-1}$.

Proof. Let $a, b \in G$. Note that

$(ab)(b^{-1}a^{-1}) = a(b(b^{-1}a^{-1})) = a((bb^{-1})a^{-1}) = a(ea^{-1}) = aa^{-1} = e$,
using the associativity of G and properties of inverses and identities.

Similarly, $(b^{-1}a^{-1})(ab) = e$. So we can conclude that $b^{-1}a^{-1}$ is the inverse of ab , that is, $(ab)^{-1} = b^{-1}a^{-1}$.

Lemma. Given $a, b, c \in G$, if $ab = ac$ then $b = c$ and if $ba = ca$ then $b = c$. (Cancellative law.)

Proof. Since G is closed under inverses, $a^{-1} \in G$. Pre-multiplying both sides of the equation $ab = ac$ gives:

$$\begin{aligned} a^{-1}(ab) &= a^{-1}(ac), \\ (a^{-1}a)b &= (a^{-1}a)c && \text{(associativity),} \\ eb &= ec && \text{(inverses),} \\ b &= c && \text{(identity).} \end{aligned}$$

Similarly, post-multiplying both sides of $ba = ca$ by a^{-1} implies $b = c$.

Cyclic groups

Groups are often thought of as objects that quantify types of symmetries. A particularly nice kind of symmetry is an isometry, that is, a transformation of a geometric object that doesn't change its shape.

Below are the isometries of a one-faced equilateral triangle:



If we call a the rotation of the triangle 120° counterclockwise, then a^2 is the rotation 240° counterclockwise (or 120° clockwise), and a^3 returns the triangle to its original position, so $a^3 = e$ in the group of these symmetries.

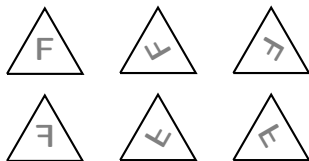
Definition. The **cyclic group** of order n , denoted C_n , is the group of rotational symmetries on a regular n -gon.

Notation. Groups can be represented in the form $\langle \text{generators} : \text{relators} \rangle$, where the generators are some elements of the group, and the relators are equations relating combinations of elements. The group is then constructed by taking all possible combinations of the generators and their inverses.

Fact. The cyclic group $C_n = \langle a : a^n = e \rangle$. We saw that $C_3 = \{e, a, a^2\}$.

Dihedral groups

Below are the isometries of a two-faced equilateral triangle, where we are now allowed to reflect the shape:



We can again call a the rotation of the triangle 120° counterclockwise, where $a^3 = e$. We now have an additional action b representing reflection, for example about the vertical altitude. Since two applications of a reflection returns the triangle to its original position, we have $b^2 = e$ in this group of symmetries.

Definition. The **dihedral group** of order $2n$, denoted D_n (or sometimes D_{2n}), is the group of isometries (rotational and reflectional symmetries) on a regular n -gon.

Fact. The dihedral group $D_n = \langle a, b : a^n = b^2 = (ab)^2 = e \rangle$. We have seen for example that $D_3 = \{e, a, a^2, b, ab, a^2b\}$.

Permutation groups

A **permutation** on n elements is any ordered arrangement of n distinct objects. We typically think of the permutations as applied to the positive integers from 1 to n , often denoted by $[n] = \{1, 2, 3, \dots, n\}$.

There are 6 permutations of $[3]$, which can be written as concatenated strings as 123, 132, 213, 231, 312, and 321.

We can also think of permutations as the **actions** that converted the string 123 to each of the different arrangements above. For example, the arrangement 213 comes from the permutation that swaps the first two positions of 123.

This can be represented by a function $\sigma : [3] \rightarrow [3]$ defined by $\sigma(1) = 2$, $\sigma(2) = 1$, and $\sigma(3) = 3$.

Notice that the permutations on $[n]$ are precisely the bijective functions on n . So the set of permutations on $[n]$ form a group under function composition.

Definition. The **symmetric group** of order $n!$, denoted S_n , is the group of all possible permutations on n objects.

Subgroups

Definition. If $(G, *)$ is a group and H is a subset of G , then H is a **subgroup** of G , written $H \leq G$, if and only if $(H, *)$ is a group.

Example. We have seen that both $\{1, -1, i, -i\}$ and $\{1, -1\}$ are groups under multiplication. Since $\{1, -1\} \subseteq \{1, -1, i, -i\}$ and $\{1, -1\}$ is a group, we therefore have that $\{1, -1\} \leq \{1, -1, i, -i\}$.

Normally to confirm a set is a group, we have to check all four axioms. But when checking for a subgroup, we can be a bit more efficient.

Lemma. (**Subgroup lemma**)

If $(G, *)$ is a group and $H \subseteq G$, then $H \leq G$ if and only if

- $H \neq \emptyset$,
- for all $a, b \in H$, we have $a * b \in H$, and
- for all $a \in H$, we have $a^{-1} \in H$.

Proof. If the listed conditions hold, then **closure** and existence of **inverses** are already satisfied. **Associativity** is inherited from the group G . Notice that since H is nonempty, we can choose some $a \in H$ along with its inverse $a^{-1} \in H$. Then $a * a^{-1} = e \in H$, showing the existence of the **identity** in H . Thus H is indeed a group, and therefore a subgroup of G .

Examples of subgroups

$$\text{Let } I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, B = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, C = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Exercise. We have already shown that $G = \{I, A, B, C\}$ is a group under matrix multiplication. Show that $H = \{I, A\}$ is a subgroup of G .

Solution. Note that H is closed under taking inverses since $I^{-1} = I \in H$, and $A^{-1} = A \in H$. Note also that H is closed under multiplication since $I^2 = A^2 = I \in H$ and $IA = AI = A \in H$. So by the subgroup lemma, $H \leq G$.

Exercise. Show that $K = \{C\}$ is **not** a subgroup of G .

Solution. Note that K is closed under taking inverses since $C^{-1} = C \in H$. However, K is **not** closed under multiplication since $C^2 = I \notin H$. So K is not a group and therefore $K \not\leq G$.

Exercise. Let $m \in \mathbb{Z}$ and set $m\mathbb{Z} = \{mx : x \in \mathbb{Z}\}$, the set of all multiples of m . Show that $(m\mathbb{Z}, +) \leq (\mathbb{Z}, +)$.

Solution. Let $a \in m\mathbb{Z}$. Then $a = mk$ for some $k \in \mathbb{Z}$, and the inverse of a is $-a = m(-k) \in m\mathbb{Z}$. So $m\mathbb{Z}$ is closed under taking inverses. Now let $b \in m\mathbb{Z}$. Then $b = m\ell$ for some $\ell \in \mathbb{Z}$, and the sum $a + b = m(k + \ell) \in m\mathbb{Z}$. So $m\mathbb{Z}$ is closed under addition. So by the subgroup lemma, $m\mathbb{Z} \leq \mathbb{Z}$.

Examples of subgroups

Here are some other examples of subgroups, which you might like to try proving as an exercise:

- For any group G , the trivial group is a subgroup – that is, $\{e\} \leq G$.
- Any group G is a subgroup of itself – that is, $G \leq G$.
- For any $n \in \mathbb{Z}^+$, we have $C_n \leq D_n \leq S_n$.
- **Definition.** The set of all $n \times n$ real matrices with determinant 1 is called the special linear group of degree n over $\mathbb{R}$, denoted $SL(n, \mathbb{R})$.

For any $n \in \mathbb{Z}^+$, we have $SL(n, \mathbb{R}) \leq GL(n, \mathbb{R})$.

- We will later (in Topic 4) define orthogonal matrices as those whose inverse equals their transpose. The real orthogonal matrices over $\mathbb{R}$ make up the set $O(n, \mathbb{R})$, and those that have determinant 1 make up the set $SO(n, \mathbb{R})$.

For any $n \in \mathbb{Z}^+$, we have $SO(n, \mathbb{R}) \leq O(n, \mathbb{R}) \leq GL(n, \mathbb{R})$ and $SO(n, \mathbb{R}) \leq SL(n, \mathbb{R})$.